

Customer Behaviour Analysis

End-to-End Data Analytics Project

Python

SQL Server

Power BI

Total Customers
5,000

Total Revenue
994K

Avg. Spend
198.84

Avg. Rating
3.63

Unique Items
30

PROJECT PIPELINE

1

Data Collection

2

EDA (Python)

3

Data Cleaning

4

SQL Analysis

5

Power BI

6

Insights

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01 EXECUTIVE SUMMARY

This report presents a comprehensive end-to-end data analytics project on customer purchasing behaviour in the retail e-commerce domain. The full pipeline spans raw data ingestion and exploratory analysis in Python, structured SQL querying in SQL Server 17.0, through to interactive visualisation in Power BI. The objective is to equip business stakeholders with actionable intelligence to improve revenue, retention, and marketing ROI.

5,000	994K	198.84	3.63	30
Total Customers	Total Revenue	Avg. Spend	Avg. Rating	Unique Items

02 ABOUT THE DATASET

Attribute	Detail
Dataset Type	Retail / E-Commerce (Customer Transaction Level)
Nature	Structured — Tabular
Record Count	5,000 customer transactions
Numerical Columns	Age, Purchase Amount (USD), Previous Purchases, Review Rating
Categorical Columns	Category, Payment Method, Season, Gender, Shipping Type
Primary Use	Customer Behaviour Analysis + Business Decision-Making

03 BUSINESS PROBLEM STATEMENT

How can a retail company understand customer purchasing behaviour to increase revenue, improve customer retention, and optimise marketing strategies?

Low Customer Retention	Customers not purchasing frequently. Key questions: Who are frequent buyers? Who are one-time buyers?
Ineffective Marketing	Discounts may not be effective. Do they increase purchases? Which customers respond to promos?
Product Demand	Which category performs best? Which products generate the highest revenue?
Payment Behaviour	Which payment methods are preferred? Are digital payments increasing?
Seasonal Demand	Does season impact sales? Which season drives maximum purchases?
Customer Segmentation	Need to identify high-value, frequent, and discount-sensitive customers.

04 DATA DICTIONARY

Column Name	Data Type	Description
customer_id	Integer	Unique identifier for each customer
age	Integer	Age of the customer
gender	Categorical	Gender (Male / Female / Other)
item_purchased	Categorical	Specific product purchased
category	Categorical	Product category (Clothing, Electronics, etc.)
purchase_amount	Float (USD)	Amount spent by the customer
location	Categorical	Customer geographic location (US State)
size	Categorical	Product size (S, M, L, XL, Free Size, N/A)
season	Categorical	Season of purchase (Winter, Summer, Fall, etc.)
review_rating	Float (1–5)	Customer satisfaction rating
subscription_status	Boolean	Subscribed (Yes / No)
shipping_type	Categorical	Shipping method selected
discount_applied	Boolean	Discount applied (Yes / No)
previous_purchases	Integer	Count of past purchases
payment_method	Categorical	Mode of payment
frequency_of_purchases	Categorical	Purchase frequency (Weekly, Monthly, etc.)

05 EXPLORATORY DATA ANALYSIS — Python (EDA)

The EDA phase was performed using Python (pandas) to load, explore, clean, and export the dataset. The following steps were executed systematically:

01	Data Loading	Imported pandas and loaded CSV using <code>pd.read_csv()</code> . Brings raw data into a structured DataFrame.
02	Initial Exploration	Used <code>df.head()</code> , <code>df.tail()</code> , <code>df.describe()</code> , <code>df.info()</code> , <code>df.isnull().sum()</code> to understand structure.
03	Data Consistency Check	Found mismatches between Item Purchased and Category; created a mapping dictionary to fix.
04	Missing Values — Size	Electronics/Accessories → 'Not Applicable'. Footwear → 'Free Size'. Clothing → mode imputation.
05	Missing Values — Rating	Used product-level mean (grouped) rather than global mean for realistic imputation.
06	Missing Values — Purchase Amount	Replaced with mean to maintain pricing consistency and avoid revenue distortion.
07	Missing Values — Previous Purchases	Replaced with 0 — implies no prior purchase; important for segmentation.
08	Duplicate Handling	Detected and removed duplicates by Customer ID — ensures no double-counting in analysis.
09	Column Standardisation	Lowercased and underscored all column names for SQL-friendliness and Power BI compatibility.
10	SQL Server Export	Installed pyodbc + sqlalchemy, connected to SQL Server, exported cleaned DataFrame to <code>Cb_Analysis</code> .

Python EDA Notebook Screenshot

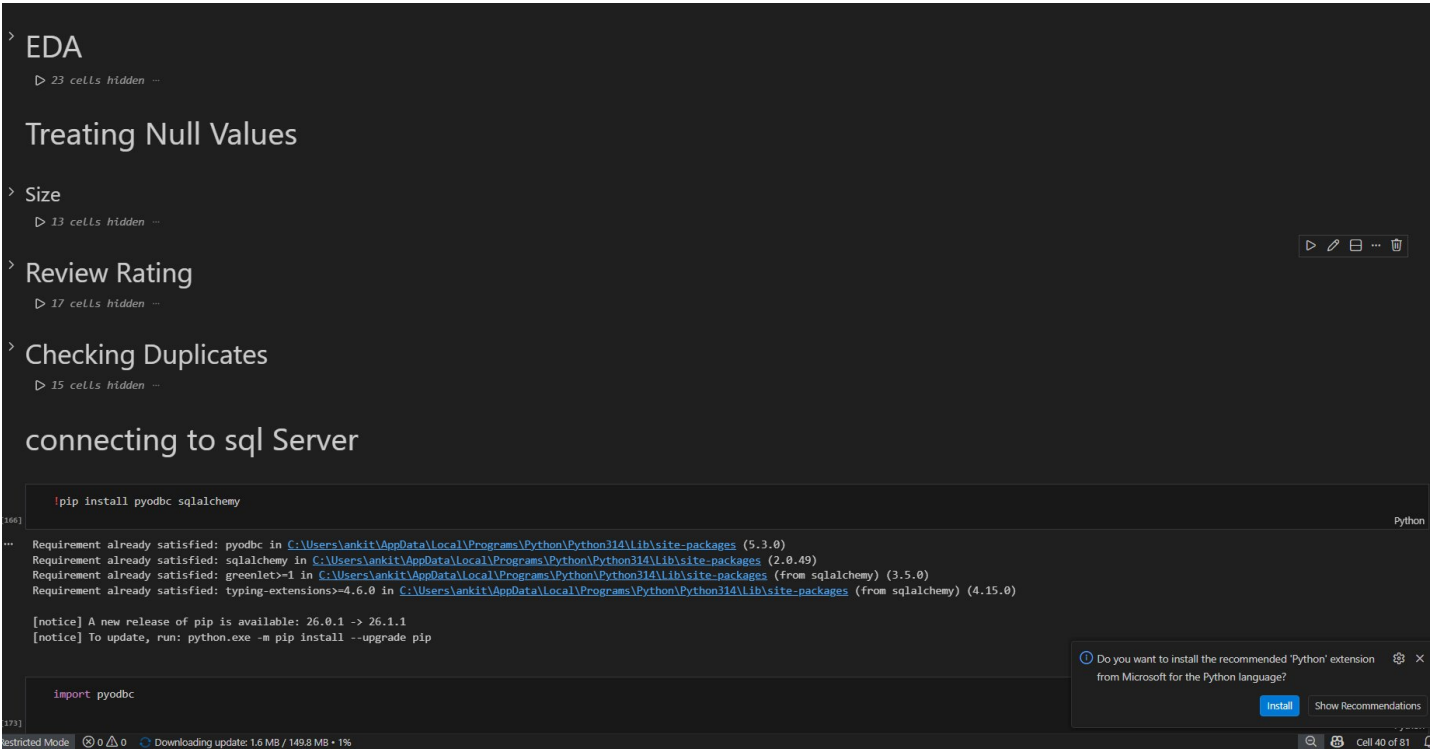


Figure 1 — VS Code Jupyter Notebook: EDA sections including Null Treatment, Size handling, Review Rating, Duplicate checking, and SQL Server connection setup.

06 DATA ANALYSIS — SQL Server

After cleaning and exporting to SQL Server 17.0, 12 business intelligence questions were answered using T-SQL against the Cb_Analysis database (dbo.customer_data table). Techniques included GROUP BY aggregations, window functions (RANK, ROW_NUMBER), CTEs, subqueries, and CASE-based segmentation.

Q1	Which category generates the highest revenue?	Electronics leads at █486,367 — 2.5× Accessories (█194,867).
Q2	Do discounts increase purchase value?	Yes — discounted avg █219.22 vs █182.24 non-discounted (20% uplift).
Q3	Revenue by gender?	Male: █390,965 Female: █352,060 Other: █251,158
Q4	High-spend discount customers?	Identified premium discount utilisers — key loyalty programme targets.
Q5	Top 5 products by rating?	Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.79)
Q6	Standard vs Express shipping?	Express: █358.12 avg (1,199 orders) vs Standard: █340.57 (1,201 orders)
Q7	Subscribed vs non-subscribed spend?	Subscribers avg █270.58 vs █165.72 — 63% higher spending!
Q8	Top discount usage products?	Laptop (51.7%), Phone (51.5%), Headphones (51.2%), Hat (50.0%), Watch (49.7%)
Q9	Customer segmentation?	Loyal: 3,085 Returning: 1,354 New: 561
Q10	Top 3 products per category?	Clothing: Shirt (315); Electronics: Phone (171); Footwear: Shoes (303)
Q11	Repeat buyers vs subscription?	1,190 repeat buyers subscribed vs 2,763 unsubscribed — large upsell gap.
Q12	Revenue by age group?	51+: █399,345 35–50: █266,707 26–35: █177,422 18–25: █150,708

SQL Server Management Studio Screenshot

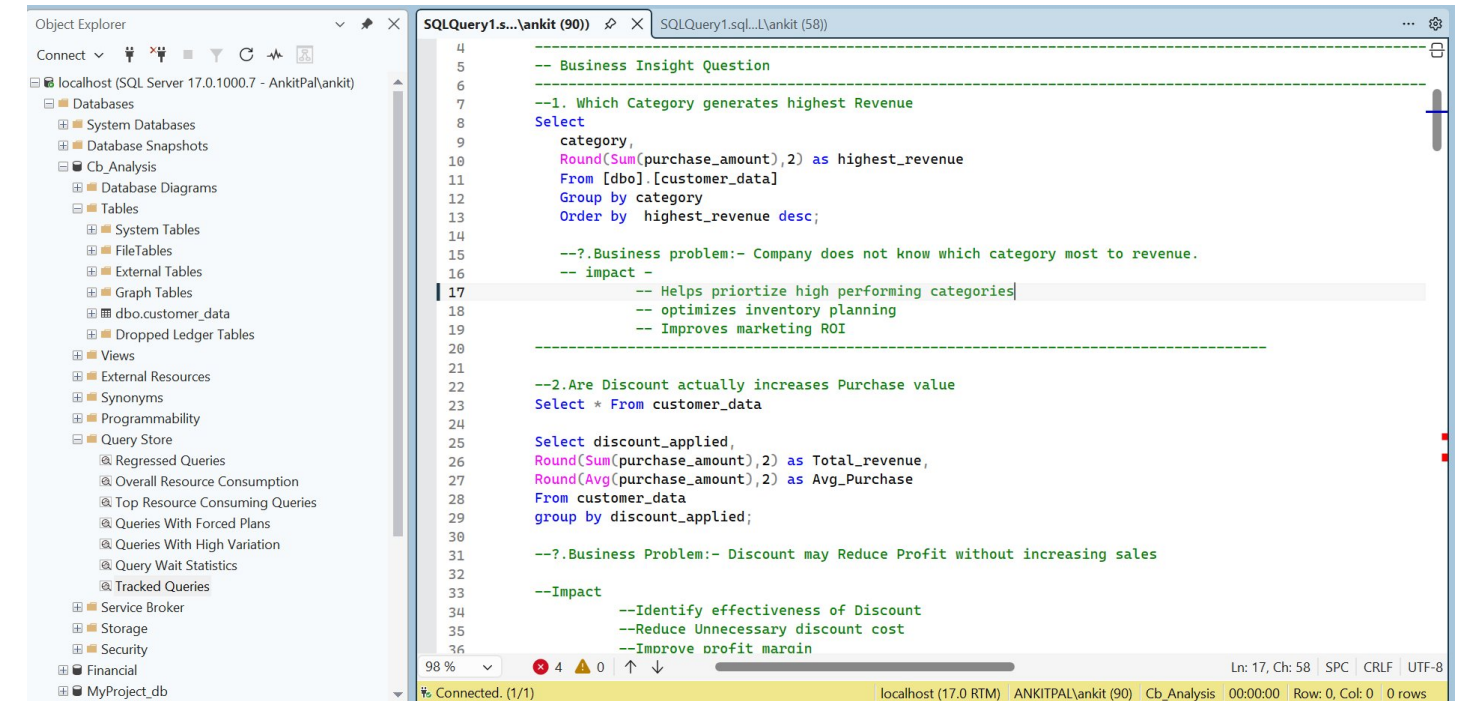


Figure 2 — SSMS: Business insight queries on Cb_Analysis database. Visible queries include category revenue, discount analysis, and gender-based revenue.

07 POWER BI DASHBOARD — Overview Page

Page 1 provides a high-level overview with interactive KPI cards, revenue breakdowns by gender, category, season, age group, payment method, shipping type, and geographic location. Interactive slicers for Category, Gender, Discount Applied, and Subscription Status allow real-time filtering across all visuals.

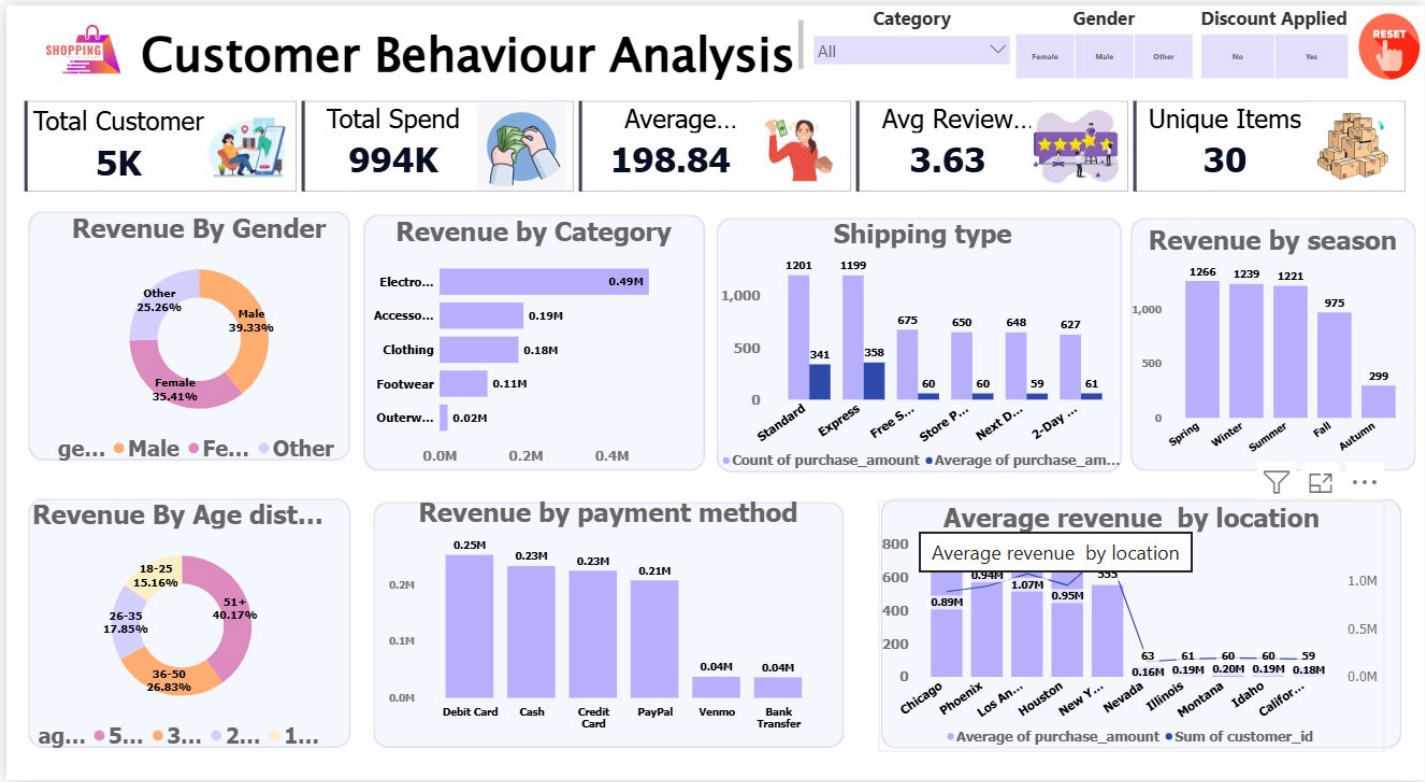


Figure 3 — Power BI Dashboard Page 1: Customer Behaviour Analysis — KPI Cards, Revenue by Gender / Category / Season / Payment Method / Shipping / Location. Slicers: Category, Gender, Discount Applied.

Dashboard Visuals Summary — Page 1

Visual	Chart Type	Key Finding
Total Customers / Spend / Rating / Items	KPI Cards	5K customers 994K revenue 3.63 avg rating
Revenue by Gender	Donut Chart	Male 39.33% Female 35.41% Other 25.26%
Revenue by Category	Horizontal Bar	Electronics dominates at 0.49M (48.9%)
Shipping Type	Clustered Bar	Standard (1,201) & Express (1,199) nearly equal
Revenue by Season	Bar Chart	Spring highest (1,266); Autumn lowest (299)
Revenue by Age Distribution	Donut Chart	51+ age group leads at 40.17%
Revenue by Payment Method	Bar Chart	Debit Card top at 0.25M (25.5%)
Avg Revenue by Location	Combo Chart	Chicago highest avg; New York highest total

08 POWER BI DASHBOARD — Detail Page

Page 2 dives into product-level performance — Top/Bottom 5 items by rating and revenue, Top/Bottom 5 locations by revenue — enabling rapid identification of star performers and underperforming SKUs across categories.

Dashboard Visuals Summary — Page 2

Visual	Chart Type	Key Finding
Top 5 Items by Rating	Horizontal Bar	Gloves (3.86), Sandals (3.84), Boots (3.82)
Bottom 5 Items by Rating	Horizontal Bar	Phone (2.94), Headphones (3.08), Watch (3.10)
Top 5 Items by Revenue	Horizontal Bar	Headphones (138K), Phone (126K), Bag (121K)
Bottom 5 Items by Revenue	Horizontal Bar	Jeans & Gloves (8K), Sneakers/Backpack/Hoodie (9K)
Top 5 Locations by Revenue	Column Chart	New York (143.9K), Los Angeles (125.7K)
Bottom 5 Locations by Revenue	Column Chart	Kansas (3.44K), Hawaii (3.75K), Florida (3.80K)

Interactive slicers — Category, Gender, Discount Applied, Subscription Status — filter all visuals simultaneously. A RESET button on the top-right restores default state for quick comparison.

09 KEY INSIGHTS & FINDINGS

Revenue & Category Insights

- Electronics leads with █486,367 revenue — nearly 2.5× the second-best category (Accessories █194,867).
- Outerwear critically underperforms at █18,524 — requires product review or targeted marketing.
- Headphones (138K), Phone (126K), and Bag (121K) are the top 3 revenue-generating products.

Customer Demographics

- The 51+ age group is the highest value segment, contributing 40.17% of total revenue.
- Male customers generate slightly more revenue (█390,965) compared to Female (█352,060).
- New York and Los Angeles lead in total location revenue; Kansas and Hawaii are lowest.

Discount & Subscription Behaviour

- Discounts increase average purchase by 20% (█219.22 vs █182.24) — strategy is effective.
- Subscribed customers spend 63% more on average (█270.58 vs █165.72) — critical CLV driver.
- 2,763 repeat buyers are NOT subscribed — this is the single largest upsell opportunity.

Shipping & Payment Patterns

- Standard and Express shipping are nearly equal (~1,200 orders each) — both must be maintained.
- Express shipping customers generate marginally higher average revenue (█358 vs █341).
- Debit Card is most popular (25.5%), Venmo (3.8%) and Bank Transfer (3.7%) lowest adoption.

Seasonal & Product Performance

- Spring drives peak volume (1,266 purchases); Autumn is critically low (299) — 78% gap.
- Top-rated products (Gloves, Sandals, Boots) are NOT in the top revenue list — rating ≠ volume.
- Phone has the lowest rating (2.94) despite being #2 in revenue — quality intervention needed.
- Loyal customers (3,085) dominate; focus should be on retention and upsell, not just acquisition.

10 BUSINESS RECOMMENDATIONS

1	Target Subscription Upsell for Repeat Buyers 2,763 repeat buyers are unsubscribed. A targeted subscription campaign could convert high-frequency shoppers, unlocking the █270+ average spend tier and significantly boosting CLV.
2	Invest Further in Electronics Category Electronics generates 48.9% of all revenue. Expand SKU range, ensure stock availability year-round, and run category-specific promotions to compound this advantage.
3	Address Product Quality for Low-Rated Electronics Phone (2.94) and Headphones (3.08) have the lowest ratings despite high revenue. Invest in QA, supplier reviews, and post-purchase feedback loops to protect long-term brand trust.
4	Launch Autumn Seasonal Campaign Autumn has only 299 purchases vs 1,266 in Spring — a 78% revenue gap. A targeted discount or limited-edition campaign in Autumn could significantly boost off-peak revenue.
5	Build Premium Loyalty Programme for 51+ Segment The 51+ group contributes 40.17% of total revenue. Design a tiered loyalty programme, premium support, and personalised recommendations specifically for this high-value demographic.
6	Incentivise Express Shipping Adoption Express shipping customers spend slightly more (█358 vs █341). Offering free/subsidised express shipping above a spend threshold could nudge average order value upward.

11 TOOLS & TECH STACK

Tool	Version / Detail	Purpose in Project
Python	pandas, numpy (Jupyter/VS Code)	Data loading, EDA, cleaning, export to SQL
SQL Server	Version 17.0.1000.7 (SSMS)	Data storage & T-SQL business intelligence queries
Power BI Desktop	Latest (2025–2026)	Interactive dashboard & visual analytics
pyodbc + sqlalchemy	5.3.0 / 2.0.49	Python to SQL Server data export pipeline
Jupyter / VS Code	Notebook environment	EDA and Python script development